

1 Transformations & families of functions

1. The graph of $f(x) = \sqrt{x}$ undergoes the following transformations in order:

- i dilation by a factor of 3 from the y axis
- ii reflection about the y axis
- iii translation of 6 units in the positive x direction

Find the equation of the transformed function g , and state its domain and range

2. Find a set of transformations that transforms the graph of $y = -\frac{3}{(2x-1)^2} + 2$ to $y = \frac{1}{x^2}$
3. State the transformations required to transform the graph of $y = \cos(x)$ to $y = -\frac{1}{5}\cos(3x - \frac{\pi}{2})$
4. State the transformations required to transform the graph of $y = 3\sin(\frac{1}{2}(x + \frac{\pi}{3})) - 5$ into the graph of $y = \sin(x)$
5. For what values of a is $y = ax^2 - 3ax + 1$ a perfect square?
6. Find the turning points of the function $f(x) = \frac{4}{5}x^5 - x^4 - \frac{28}{3}x^3 + 2x^2 + 24x + 19$
7. Let $f(x) = (x-1)^2(x-2) + 1$
 - (a) Find the values for p for which the equation $f(x) = p$ has exactly one solution
 - (b) Find the values for h for which only one of the solutions to $f(x+h) + 1 = 0$ is positive
8. Find a set of transformations that transforms the graph of $y = 2^x$ to $y = 3^x$
9. Find values of p such that the function $f(x) = px^2 + 13x + p$ and $y = 12x + 3p$ only has one intersection